

ON THE MINIMA OF THE TIME-INTEGRATED PERTURBATION
FACTOR IN THE SCHERER-BLUME THEORY[†]

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The minima in the correlation time dependence of the Scherer-Blume time-integrated attenuation coefficients for the hyperfine perturbation of ions recoiling in gas are studied. Its position and depth are determined for different physical situations and comparison with experimental data is shown.

1. Introduction

Measurements of γ - γ angular correlations perturbed by hyperfine interactions experienced by excited nuclei recoiling in gas are used extensively for the determination of magnetic moments of nuclear levels [1], as well as for the investigation of the atomic processes involved [2]. The measured attenuation coefficients are usually compared with those evaluated using the theory developed by Scherer [3] and Blume [4] based on a stochastic perturbation model.

A striking feature in the behaviour of the perturbation factor is the observed minimum in the correlation time dependence of the attenuation coefficients [1, 2, 7]. Remarkably enough, the existence of this minimum is predicted by the Scherer-Blume theory although its physical meaning remains obscure.

Some approximate conditions for the existence of the minimum have been derived by Zemel and Niv [5] for the correlation time dependence of the time differential coefficient ($G_K(t)$). They show that the position of the minimum is directly proportional to the g -factor of the nucleus and its depth is a function of the time the hyperfine fields act on the nucleus.

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The purpose of the present note is to study the dependence of the time-integrated coefficient ($\overline{G_K}(\infty)$) with the correlation time in an attempt to gain some understanding about the influence of the atomic process involved in the recoiling of excited nuclei in gas. In particular, we focussed our attention on the properties of the minima of $\overline{G_K}(\infty)$, looking for existence conditions and simplified expressions describing its positions and depth.

2. Results

Our starting point is the expression for the time-integrated perturbation coefficients obtained by Blume [4]:

$$\overline{G_K}(\infty) = \left[\sum_{\nu} \frac{1}{[C_{\nu}/(1+\tau/\tau_c + i\omega_{\nu}\tau)] - \tau/\tau_c} \right]^{-1}. \quad (1)$$

The frequencies ω_{ν} are given by $\omega_{\nu} = \omega_{FF'} = (E_F - E_{F'})/\hbar$, where E_F and $E_{F'}$ are the eigenvalues of the hyperfine interactions hamiltonian.

The coefficients C_{ν} are products of a geometrical term, depending on the atomic angular momentum J , the nuclear spin I , the total angular momentum F , and a statistical term specifying the charge state distribution and the probability of occupation of a certain configuration within this charge state [6]. They are real numbers satisfying

$$C_{\nu} = C_{-\nu} \quad \text{and} \quad \sum_{\nu} C_{\nu} = 1. \quad (2)$$

If we are looking for a minimum of $\overline{G_K}(\infty)$ when τ_c varies, we have to solve

$$\left. \frac{\partial \overline{G_K}(\infty)}{\partial \tau_c} \right|_{\tau_c = \tau_c^*} = 0, \quad (3)$$

where τ_c^* 's are the values of τ_c which minimize (1).

Using eqs. (1)-(3), we get

$$\sum_{\nu} \frac{C_{\nu}}{(1+\tau/\tau_c^* + i\omega_{\nu}\tau)^2} = \left(\sum_{\nu} \frac{C_{\nu}}{1+\tau/\tau_c^* + i\omega_{\nu}\tau} \right)^2. \quad (4)$$

We now introduce the "effective mean-life" τ_{eff} :

$$\frac{1}{\tau_{eff}} = \frac{1}{\tau} + \frac{1}{\tau_c}, \quad (5)$$

which, for "pure" static ($\tau \ll \tau_c$) interactions reduces to the nuclear mean life τ , thus controlling the position of the minimum. On the other hand, for highly fluctuant hyperfine fields ($\tau \gg \tau_c$), $\tau_{eff} \approx \tau_c$ and the correlation time dominates the behaviour of the perturbation factor.

Using the fact that $\omega_{\nu} = -\omega_{-\nu}$, we then obtain the following

equation in terms of τ_{eff} :

$$C_0 + \sum_{\nu \neq 0} C_\nu \frac{1 - \omega_\nu^2 \tau_{\text{eff}}^{*2}}{[1 + \omega_\nu^2 \tau_{\text{eff}}^{*2}]^2} = \left\{ C_0 + \sum_{\nu \neq 0} \frac{C_\nu}{1 + \omega_\nu^2 \tau_{\text{eff}}^{*2}} \right\}^2. \quad (6)$$

The solutions of this equation give the values of $\tau_{\text{eff}} = \tau_{\text{eff}}^*$ (and consequently of τ_c^*) for which $\overline{G_K}(\infty)$ shows a minimum. We note that the solutions are not always real: for instance, when $\omega_\nu \tau_{\text{eff}}^* < 1$ for all positive ν , $\overline{G_K}(\infty)$ does not exhibit minima. It is not possible in general to find an analytical solution of eq.(6) and therefore it will be useful to consider the following different situations: (i) systems with three hyperfine frequencies, one of which is zero and the other two equal in magnitude, but with opposite signs; (ii) systems with several frequencies very different from one another; (iii) systems where these frequencies are approximately the same.

2.1. Only one frequency with modulus different from zero

When there are only two possible values of F , one has necessarily $\omega_0 = 0$ and $\omega_1 = -\omega_{-1} \equiv \omega$ and eq.(6) is easily solved, giving just one solution

$$\tau_{\text{eff}}^* = \frac{1}{\omega \sqrt{C_0}}, \quad (7)$$

or, in terms of the correlation time:

$$\tau_c^* = \frac{\tau}{\omega \tau \sqrt{C_0} - 1}. \quad (8)$$

Since τ_c^* must be positive, the condition for the existence of a minimum turns out to be $\omega \tau \sqrt{C_0} > 1$. Taking $\sqrt{C_0} \approx 1$ (as is usually the case) eq.(8) can be written as:

$$\tau_c^* = \frac{\tau}{\omega \tau - 1}. \quad (9)$$

One can see that if $\omega \tau \gg 1$, i.e. on the average the nuclei precess several times before the de-excitation process takes place, then the following relation holds:

$$\tau_c^* \approx \frac{1}{\omega}, \quad (10)$$

showing that the minimum of $\overline{G_K}(\infty)$ occurs when the frequency of atomic collisions coincides approximately with the hyperfine frequency.

We also note that in this situation the position of the minimum is not sensible to the probability of occupying states with different values of F .

Another relevant quantity is the minimum value of $\overline{G_K}(\infty)$. Substitution of eq.(8) into eq.(1) gives:

$$\overline{G}_K(\omega) \Big|_{\tau_c} = \tau_c^* = \left[1 + \frac{C_1}{\sqrt{C_0}} \omega \tau \right]^{-1} \quad (11)$$

Equation (11) shows that while the position of the minimum of $\overline{G}_K(\omega)$ depends uniquely on ω , the depth of this minimum is a function of the coefficient C_V which is directly related to the cross sections of the atomic interactions involved. It will be seen below that the product $C_V \omega_V$ also plays an important role when several frequencies are present.

We can also note from eq. (11) that $\overline{G}_K(\omega)$ cannot be zero. This is simply a consequence of the fact that the nuclei are initially aligned and, irrespective of how strong the perturbation is, the average-in-time alignment will never be exactly zero.

Figure 1a shows the shape of the attenuation coefficients $\overline{G}_2(\omega)$, for this particular case, as a function of τ/τ_c for a set of relevant parameters.

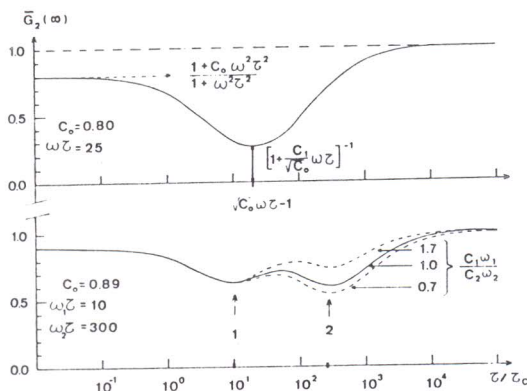


Fig. 1. Upperpart: prediction of $\overline{G}_2(\omega)$ for only one frequency with modulus different from zero. Lowerpart: the existence of minima in $\overline{G}_2(\omega)$ is a consequence of the very different values of the two assumed frequencies. One can note that when $C_1 \omega_1 = C_2 \omega_2$ the depth of two minima are the same.

2.2. Several frequencies: $|\omega_1| \gg |\omega_2| \gg |\omega_3| \dots$

Although an analytical solution of eq. (6) cannot be reached, numerical calculations of eq. (1) for particular cases can show relevant features of the correlation time dependence of the attenuation coefficients. When all hyperfine frequencies involved are quite different from one another, numerical analysis shows the existence of several minima on $\overline{G}_K(\omega)$, each of them with characteristics roughly equal to those corresponding to a $\overline{G}_K(\omega)$ with just one specific frequency. In other words, the shape of $\overline{G}_K(\omega)$ corresponding to several frequencies can be approximately obtained by supposing the shapes of $\overline{G}_K(\omega)$'s corresponding to a single $|\omega_V|$.

Figure 1b illustrates these features as well as the influence of

$$\tau_c^* = \frac{1}{\bar{\omega}}. \quad (17)$$

Note that this equation is formally identical to eq.(10) (which corresponds to the case of only one non-zero frequency ω). The only change when several similar frequencies are present is that the average $\bar{\omega}$ appears instead of ω .

If $\omega_v \tau_{eff} \approx 1$ for all v , the assumption $x_0 \neq 1$ is not satisfied in eq.(15). A similar derivation however gives $\tau_{eff}^* \bar{\omega}^2 = 3/(2+C_0)$ and therefore eq.(17) remains unchanged.

The value of $\overline{G_K}(\infty)$ in the minimum can now be obtained by substituting eq.(17) into eq.(1):

$$\overline{G_K}(\infty) \Big|_{\tau_c} = \tau_c^* = \left[1 + \frac{1-C_0}{2C_0} \frac{\tau}{\tau_{eff}^*} \right]^{-1} = \left[1 + \frac{1-C_0}{2\sqrt{C_0}} \bar{\omega} \tau \right]^{-1}. \quad (18)$$

The similarity between equations (18) and (12) shows again that the effect on $\overline{G_K}(\infty)$ of several hyperfine frequencies, which are not quite different from one another, is equivalent to the effect of an average frequency $\bar{\omega}$.

Finally, it is interesting to note that eq.(18) can be alternatively obtained in the approximation $C_0 \approx 1$ by taking the time average of $G_K(t, \tau_c^*)$ given in ref. [5].

3. Discussion and applications

The equation for $\overline{G_K}(\infty)$ which we have used, [eq.(1)], assumes that τ_c is maintained constant during the de-excitation of the nuclear state. Therefore, to compare our results with experimental values, we have to search for a physical system satisfying this condition. An example is found in the decay of ^{127}Xe into ^{127}I in gaseous sources through the 203 keV level ($\tau = 0.55$ ns).

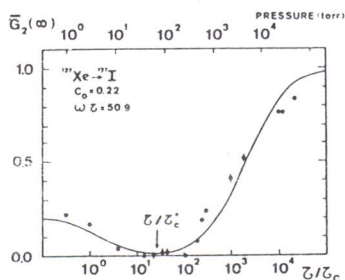


Fig. 2. Pressure dependence of the 172-203 keV angular correlation in ^{127}Xe . The continuous line is the prediction of the Scherer-Blume theory.

Figure 2 shows a comparison between experimental data [7] and our theoretical results. The abscissa τ/τ_c was obtained taking $\tau_c = 1/nv\sigma$, where n is the number of molecules per cm^3 , v is the thermal velocity

of the ions (2.5×10^4 cm/s) and σ is the total atomic cross sections ($808 \pm 234 \times 10^{-15}$ cm²) [8]. Thus, the relation between τ_c and the pressure for this case is τ_c (ns) = $1.4/P$ (torr).

Considering now the existence of only one minimum, at 60 torr, in the $\bar{G}_2(\omega)$ dependence with pressure, we can use the results obtained in sect. 2 to calculate C_0 and $\bar{\omega}$. For the present purposes, these calculations were performed in the scope of just one frequency with modulus different from zero.

The obtained values are: $C_0 = 0.22$ and $\bar{\omega} = 92.7$ GHz, (which satisfy the condition $\omega \tau \sqrt{C_0} > 1$ for the existence of a minimum).

Using these values, one can now compute the theoretical $\bar{G}_2(\omega)$ which turns out to be in very good agreement with the experimental data in the whole range of τ_c .

Thus, our analysis shows that, although the Scherer-Blume stochastic model relies on a considerable amount of limitative assumptions, it nevertheless reproduces fairly well the actual experimental data in the whole range of τ_c .

Concerning the approximate expressions that we have obtained, they have shown to be reasonably precise as to be employed in the study of fluctuating hyperfine fields when the presence of too many frequencies makes the use of the exact equation very "complicated". In particular, they provide a simple way of restricting the τ_c region where the minimum of the attenuation coefficient occurs.

It is interesting to note that the approximate expressions given by eq. (10) and eq. (17) show that the experimental determination of the position of the minima gives the value of the hyperfine frequency ω ($\bar{\omega}$) and hence the magnetic field acting on the nucleus. On the other hand, the knowledge of the depth of the minima provides information about the occupation probability of the atomic states.

As we stated above, no satisfactory explanation of the existence of the minimum has been found. It is only known [9] that solely "fluctuating orientation" models reproduce this property of the attenuation factor which has been found experimentally. Although no general conclusions may be drawn from our computations, they indicate that the value of τ_c at which it occurs is directly related to an average of the hyperfine frequencies weighted with the coefficients C_V , related to the charge state distribution and occupation probabilities of the hyperfine levels.

We end by noting that our computations show that for the case of several frequencies quite different from one another, the existence of more than one minimum is to be expected in $\bar{G}_2(\omega)$. In this case (which, to our knowledge, has not yet been found experimentally) the shape of the curve should be a superposition of $\bar{G}_K(\omega)$'s, each one corresponding to a single hyperfine frequency ω_V .

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